

Justin Lo

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EDUCATION

B.S. Computer Science | California State University, Bakersfield

May 2026

B.S. Biology, Concentration in Biotechnology | California State University, Bakersfield

TECHNICAL SKILLS

Languages: C++, CUDA, Python, Java, C#, SQL (Postgres, MariaDB, MySQL, SQLite), TypeScript/JavaScript, HTML/CSS

APIs: OpenGL, Vulkan, Unity, OpenXR

Tools & Platforms: Git, GitLab CI/CD, AWS (Lambda, Step Functions, ECS, Bedrock), Docker

Frameworks & Libraries: React, Next.js, Node.js, Vue, JUnit

EXPERIENCE

XR/VR Research Assistant | California State University, Bakersfield

January 2026 – Present

- Evaluated co-located world-space alignment strategies including manual anchor placement and periodic spatial refresh, ultimately integrating Meta Spatial Anchors via the OpenXR extension to minimize inter-device drift
- Optimized GitLab CI/CD pipeline with Unity Library caching and parallelized build targets, reducing average build time by 25% across Meta Quest and Android XR configurations
- Reorganized Unity project layout and build configuration, migrated hardcoded API secrets to GitLab Secrets Manager to prevent security concerns

Data Analysis and Qualitative Insights Extern | Beats by Dre.

February 2026 – May 2026

- Performed sentiment analysis on Beats by Dre. consumer reviews using Python, pandas, and natural language processing (NLP) libraries, identifying semantic patterns in brand perception across product lines
- Conducted exploratory data analysis on large consumer datasets to surface behavioral trends, visualized findings to support data-driven brand-building recommendations
- Leveraged Gemini AI to summarize and interpret qualitative feedback at scale, synthesizing insights into strategic deliverables for product and marketing use cases

Instructional Student Assistant | California State University, Bakersfield

January 2025 – May 2026

- Support 30+ students across Programming I and Software Engineering courses, reinforcing foundational programming concepts and software development practices including Agile, JUnit, and CI/CD with Jenkins
- Delivered lecture on Dependency Injection to Software Engineering cohort; developed supplementary material on Git branching strategies and version control workflows
- Assist students with DSA as a department tutor, providing structured problem-solving guidance during drop-in sessions

Software Developer | California Energy Research Center

September 2025 – March 2026

- Built and maintain two public-facing websites for the California Energy Research Center and its annual conference within Omni content management system (CMS), serving CSUB students, faculty, and industry partners including Chevron and Edison International
- Developed custom JavaScript components and CSS animations within the CMS asset source editor
- Integrated third-party calendar and email APIs to support conference event registration and scheduling, extending CMS functionality beyond out-of-the-box capabilities

Software Engineer Intern | Aethor Systems

June 2025 – October 2025

- Developed bare-metal C++ firmware for a closed-loop liquid cooling system on TI MCUs, integrating thermistor sensing, pulse-width modulation (PWM)-driven fan/pump actuation, and threshold-based thermal control for defense equipment
- Applied the Steinhart-Hart equation to linearize thermistor ADC readings across -40°C to 105°C, reducing temperature error by 0.5°C over linear approximation across 4 channels
- Authored serial peripheral interface (SPI) and universal asynchronous receiver-transmitter (UART) peripheral drivers from register level on TI MCUs and implemented a real-time RTT telemetry pipeline, logging timestamped sensor readings from 4 channels at 10ms intervals for post-hoc thermal diagnostics

STEAM Camp Mentor | Kern County Superintendent of Schools

June 2023 – June 2024

- Led hands-on STEAM activities for elementary and middle school students
- Mentored students through interactive experiments, projects, and problem-solving activities in small groups
- Collaborated with fellow mentors and KCSOS staff to organize activities, manage groups, and maintain lesson schedules
- Adapted activities to different age groups and encouraged creativity, teamwork, and curiosity

PROJECTS

Travyl | Next.js, React, TypeScript, AI

- Built an AI-native travel planning platform with personalized itinerary generation and collaborative trip planning
- Provisioned cloud infrastructure with Terraform across AWS and GCP, using infrastructure-as-code for reproducible deployments
- Developed a real-time itinerary generation service using WebSockets, streaming AI-generated travel recommendations and updates to connected clients